Colette Suhjung Koo

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Education

- **Korea Advanced Institute of Science and Technology (KAIST)** — Daejeon, Korea
  M.S. in AI Transformation (AX) — *September 2026 – Present*
- **Konkuk University** — Seoul, Korea
  Bachelor of Science - Computer Science and Engineering; GPA: 3.77/4.5 — *March 2023 - February 2025*
  Courses: *Computer Graphics, Advanced Computer Graphics, Human Computer Interaction, Digital Image Processing, Computer Vision*
- **Sungshin Women's University** *(Transferred)* — Seoul, Korea
  Majored in AI Convergence, Artificial Intelligence; GPA: 4.2/4.5 — *March 2021 - February 2023*

Research Interest

- **3D Facial Reconstruction & Generation & Modeling**
- **Facial Retargeting & Facial Animation Transfer**
- **Neural Rendering & Computer Graphics**

Publications

- **FaceJEPA: Canonicalizing 3D Faces for Self-Supervised Representation Learning**: Kwan Yun, **Colette Suhjung Koo**, and Junyong Noh, *ACM Transactions on Graphics (TOG)(Under Review).*
- **EDGAR : Efficient Diffusion-Driven Gaussian Avatar Refinement via Synthetic Failure**: Kwan Yun, Minhyeok Bang, **Colette Suhjung Koo**, and Junyong Noh, *ACM Transactions on Graphics (TOG)(Under Review).*
- **X-AVDT: Audio-Visual Cross-Attention for Robust Deepfake Detection**: Youngseo Kim, Kwan Yun, Seokhyeon Hong, Sihun Cha, **Colette Suhjung Koo**, and Junyong Noh, *CVPR 2026(Forthcoming).*
- **Realistic Lighting Implementation and Optimization Using Physically Based Rendering and Image-Based Lighting**: **Suhjung Koo** and Hyungseok Kim, *HCI KOREA 2025*, pp. 1022–1026.

Research Experience

- **Visual Media Lab, KAIST** — Adviser: Junyong Noh
  Research intern — *July 2025 - August 2026*
  - **FaceFeat3D**: ACM Transactions on Graphics (TOG)(Under Review)
    Developing a self-supervised 3D face representation model leveraging MAE paradigm and curating a large-scale 3D human face mesh dataset.
  - **EDGAR++**: ACM Transactions on Graphics (TOG)(Under Review)
    Developing a conditional diffusion framework to refine Gaussian head avatars by correcting artifacts, and constructing EDGAR-SET, a specialized dataset of synthetic failure cases for robust identity preservation.
    - Utilized 3D Gaussian Splatting to synthesize EDGAR-SET, and conducted experiments to improve novel-view rendering of 3D Gaussian avatars in sparse-view scenarios by leveraging EDGAR-refined high-fidelity data.
  - **X-AVDT**: CVPR 2026(Forthcoming)
    Introduced X-AVDT and the MMDF benchmark, advancing robust audio–visual deepfake detection by probing internal diffusion signals for cross-generator generalization.
    - Conducted experiments to evaluate both the X-AVDT model and the MMDF dataset through comprehensive quantitative comparisons, demonstrating the model's SOTA performance and the dataset's efficacy as a robust benchmark.
- **Virtual Reality Lab, Konkuk University** — Adviser: Hyungseok Kim
  Undergraduate research assistant — *July 2024 - February 2025*
  - Developed a high-performance rendering pipeline using modern OpenGL, achieving real-time visualization of high-density models with 10M+ polygons and massive point cloud data.
  - Implemented Physically Based Rendering (PBR) and Image-Based Lighting (IBL) systems by developing adaptive texture formulas and modular shaders, utilizing importance sampling and refined microfacet models to achieve photorealistic high-frequency reflections.
  - Developed functionality to load multiple GLTF models with a system that assigns optimized shaders and rendering parameters at load time to meet the specific rendering requirements of each model.
  - Presented a poster on realistic lighting optimization at the HCI KOREA 2025 Conference.
- **School of AI Convergence, Sungshin Women's University** — Adviser: Haewon Byun
  Undergraduate research assistant — *January 2022 - June 2022*
  - Gained hands-on experience in processing real-world datasets and optimizing machine learning models to improve performance and accuracy by solving Kaggle problems.
  - Analyzed research papers and expanded concepts of neural networks and machine learning models.

Skills

- **Languages**: English: TOEFL iBT Score 103/120
  Korean: Native
- **Programming Languages**: Python, C++, GLSL, Kotlin, Java
- **Frameworks**: OpenGL, OpenCV, PyTorch, TensorFlow, Keras, DirectX, Vulkan

Awards and Scholarships

- **Research For Undergraduate Students (RUS) Program Scholarship**: 2024
  'Real-Time Realistic Rendering Technology for Large-Scale Digital Twin', Konkuk University
- **Wrtn Technologies Inc. 2nd Ideathon, Generative AI Ideathon On Konkuk**: Excellence Award 2024
  'Petamong: Helping Young Adults Achieve Daily Goals with a Growing Pet Companion'
- **STARTUP021 2023 Entrepreneurship Club Grant**: 2023
  Funding for 'Campus Navigation System and AI Chatbot Service'
- **Software Innovation Competition**: Gold Prize 2022
  'CAMVI: Campus Navigation System', Sungshin Women's University
- **S+ Mileage Scholarship**: 2022
  Awarded for outstanding participation in school activities, Sungshin Women's University
- **Scholarship for Academic Excellence**: Top performance: 2022 (2nd Yr, 1st Sem), 2021 (Full Year) 2021–2022
- **Language Skills Scholarship**: 2021
  Awarded for excellent English proficiency score

Projects

- **Graduation Project: Conference Record Management Service Using Computer Vision**:
  - Used OCR and STT in developing a conference record management service which summarizes and records meetings, and further made the sharing of notes and schedules possible with team members.
  - Achieved a recognition accuracy of 95% by fine-tuning EasyOCR model with approximately one million handwritten Korean text images.
  - Showcased the project's key features and impact by performing a one-minute madness pitch and organized a metaverse online exhibition.
- **'Nanuming' Project: A Non-Face-to-Face Sharing Service for Childcare Supplies**:
  - Implemented a non-face-to-face childcare supply sharing service using the 'Nanuming Locker' developed by 3D printing and Arduino.
  - Developed an AI-based category recommendation system based on titles and implemented an image similarity analysis system utilizing Google Cloud's Vertex AI generative models and LangChain.
  - Conducted actual user tests and feedback for service improvement and verification.

Extracurricular Activities

- **Google Developer Student Clubs at Konkuk University**: Regular Member of AI October 2023 - July 2024
  - **Google Solution Challenge**: Led the 'Nanuming' project, a non-face-to-face sharing service for childcare products using a self-designed 'Nanuming Box.'
  - **AI Competition**: Achieved top 21% ranking with fellow AI team members in an AI competition on Dacon to predict prices of specialty products from Jeju.
  - **Presentation**: Gave a presentation on documentation, organization, and planning, focusing specifically on the Bullet Journal method.

References

- **Junyong Noh**
  Professor, Graduate School of Culture and Technology, KAIST
  - **Contact**: Email: junyongnoh@kaist.ac.kr
- **Hyungseok Kim**
  Professor, Dept. of Computer Science and Engineering, Konkuk University
  - **Contact**: Email: hyuskim@konkuk.ac.kr